WORKED SOLUTION

STEP III 2025, Question 1

Pure • Beta Function Integrals

STEP III 2025, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Everything here is a substitution or the identity $x + (1 - x) = 1$. Reflecting $x \mapsto 1 - x$ swaps the roles of p and q; splitting the integrand with $x + (1 - x)$ relates neighbouring values; $x = \sin^2\theta$ turns the integral into a trigonometric one; and $x = t/(1+t)$ stretches $[0, 1)$ onto $[0, \infty)$. The final integral is then a chain of the earlier results, ending at an integral of $\sin^4\theta$ that we can do by hand.

Solution

Part (i)

Substitute $u = 1 - x$, so $\mathrm{d}x = -\mathrm{d}u$, and $x = 0, 1$ correspond to $u = 1, 0$:

$$\mathrm{b}(p, q) = \int_1^0 (1-u)^{p-1} u^{q-1} \, (-\mathrm{d}u) = \int_0^1 u^{q-1}(1-u)^{p-1} \, \mathrm{d}u = \mathrm{b}(q, p).$$

Part (ii)

Adding the two integrals on the right-hand side,

$$\begin{aligned} \mathrm{b}(p+1, q) + \mathrm{b}(p, q+1) &= \int_0^1 x^{p-1}(1-x)^{q-1}[x + (1-x)] \, \mathrm{d}x \\ &= \int_0^1 x^{p-1}(1-x)^{q-1} \, \mathrm{d}x = \mathrm{b}(p, q), \end{aligned}$$

which rearranges to $\mathrm{b}(p+1, q) = \mathrm{b}(p, q) - \mathrm{b}(p, q+1)$.

Now put $q = p$. By part (i), $\mathrm{b}(p, p+1) = \mathrm{b}(p+1, p)$, so

$$\mathrm{b}(p+1, p) = \mathrm{b}(p, p) - \mathrm{b}(p+1, p) \quad \Longrightarrow \quad \mathrm{b}(p+1, p) = \tfrac{1}{2}\,\mathrm{b}(p, p).$$

Part (iii)

Let $x = \sin^2\theta$, so $\mathrm{d}x = 2\sin\theta\cos\theta\,\mathrm{d}\theta$ and $1 - x = \cos^2\theta$. As θ runs from 0 to $\frac{1}{2}\pi$, x increases from 0 to 1, and both $\sin\theta$ and $\cos\theta$ are non-negative, so $x^{p-1} = (\sin\theta)^{2p-2}$ and $(1-x)^{q-1} = (\cos\theta)^{2q-2}$ with no sign trouble. Hence

$$\mathrm{b}(p,q) = \int_0^{\frac{1}{2}\pi} (\sin\theta)^{2p-2}(\cos\theta)^{2q-2}\cdot 2\sin\theta\cos\theta\,\mathrm{d}\theta = 2\int_0^{\frac{1}{2}\pi} (\sin\theta)^{2p-1}(\cos\theta)^{2q-1}\,\mathrm{d}\theta.$$

For the second result, take $q = p$ and use $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$:

$$\mathrm{b}(p,p) = 2\int_0^{\frac{1}{2}\pi} \left(\tfrac{1}{2}\sin 2\theta\right)^{2p-1}\mathrm{d}\theta = 2^{2-2p}\int_0^{\frac{1}{2}\pi} (\sin 2\theta)^{2p-1}\,\mathrm{d}\theta = 2^{1-2p}\int_0^{\pi} (\sin\phi)^{2p-1}\,\mathrm{d}\phi,$$

where $\phi = 2\theta$.

> **Ask yourself** How do we get back to an integral over $[0, \frac{1}{2}\pi]$ that looks like $\mathrm{b}(p,\cdot)$?

The graph of $\sin\phi$ on $[0,\pi]$ is symmetric about $\phi = \frac{1}{2}\pi$ (substitute $\phi \mapsto \pi - \phi$), so the integral over $[0,\pi]$ is twice the integral over $[0, \frac{1}{2}\pi]$. Also, the trigonometric form with $q = \frac{1}{2}$ has $(\cos\theta)^{2q-1} = (\cos\theta)^0 = 1$, giving

$$\mathrm{b}(p,\tfrac{1}{2}) = 2\int_0^{\frac{1}{2}\pi} (\sin\theta)^{2p-1}\,\mathrm{d}\theta.$$

Therefore

$$\mathrm{b}(p,p) = 2^{1-2p}\cdot 2\int_0^{\frac{1}{2}\pi} (\sin\phi)^{2p-1}\,\mathrm{d}\phi = 2^{1-2p}\,\mathrm{b}(p,\tfrac{1}{2}) = \frac{1}{2^{2p-1}}\,\mathrm{b}(p,\tfrac{1}{2}),$$

as required.

Part (iv)

We want x to go from 0 to 1 as t goes from 0 to ∞, and $1 - x$ to be simple; $x = \dfrac{t}{1+t}$ does both, with

$$1 - x = \frac{1}{1+t}, \qquad \mathrm{d}x = \frac{(1+t) - t}{(1+t)^2}\,\mathrm{d}t = \frac{\mathrm{d}t}{(1+t)^2}.$$

As t increases from 0 to ∞, x increases from 0 towards 1. So

$$\mathrm{b}(p,q) = \int_0^{\infty} \frac{t^{p-1}}{(1+t)^{p-1}}\cdot\frac{1}{(1+t)^{q-1}}\cdot\frac{1}{(1+t)^2}\,\mathrm{d}t = \int_0^{\infty}\frac{t^{p-1}}{(1+t)^{p+q}}\,\mathrm{d}t.$$

Part (v)

Match the integrand with part (iv): $p-1=\frac{3}{2}$ and $p+q=6$, so $p=\frac{5}{2}$, $q=\frac{7}{2}$. The integral is $\mathrm{b}(\frac{5}{2},\frac{7}{2})$, and we reduce it step by step:

$$\begin{aligned}\mathrm{b}(\tfrac{5}{2},\tfrac{7}{2}) &= \mathrm{b}(\tfrac{7}{2},\tfrac{5}{2}) && \text{by (i)}\\ &= \tfrac{1}{2}\,\mathrm{b}(\tfrac{5}{2},\tfrac{5}{2}) && \text{by (ii) with } p=\tfrac{5}{2}\\ &= \tfrac{1}{2}\cdot\frac{1}{2^4}\,\mathrm{b}(\tfrac{5}{2},\tfrac{1}{2}) && \text{by (iii) with } p=\tfrac{5}{2}.\end{aligned}$$

Finally, from the trigonometric form with $p=\frac{5}{2}$, $q=\frac{1}{2}$,

$$\mathrm{b}(\tfrac{5}{2},\tfrac{1}{2}) = 2\int_0^{\frac{1}{2}\pi}\sin^4\theta\,\mathrm{d}\theta.$$

Using $\sin^2\theta=\frac{1}{2}(1-\cos 2\theta)$ twice,

$$\sin^4\theta=\tfrac{1}{4}(1-2\cos 2\theta+\cos^2 2\theta)=\tfrac{1}{8}(3-4\cos 2\theta+\cos 4\theta).$$

The cosine terms integrate to zero over $[0,\frac{1}{2}\pi]$ (their antiderivatives $\sin 2\theta$ and $\sin 4\theta$ vanish at both ends), so $\int_0^{\frac{1}{2}\pi}\sin^4\theta\,\mathrm{d}\theta=\frac{3}{8}\cdot\frac{1}{2}\pi=\frac{3\pi}{16}$ and $\mathrm{b}(\frac{5}{2},\frac{1}{2})=\frac{3\pi}{8}$. Hence

$$\int_0^\infty \frac{t^{\frac{3}{2}}}{(1+t)^6}\,\mathrm{d}t=\frac{1}{32}\cdot\frac{3\pi}{8}=\frac{3\pi}{256}.$$

Checks

- With $p=q=1$ the definition gives $\mathrm{b}(1,1)=1$, and (ii) gives $\mathrm{b}(2,1)=\frac{1}{2}$, which matches $\int_0^1 x\,\mathrm{d}x$ directly.
- Also with $p=1$, part (iii) says $\mathrm{b}(1,1)=\frac{1}{2}\,\mathrm{b}(1,\frac{1}{2})=\frac{1}{2}\int_0^1(1-x)^{-1/2}\,\mathrm{d}x=\frac{1}{2}\cdot 2=1$. Consistent.
- Numerically, $3\pi/256\approx 0.036816$, and a direct numerical integration of $t^{3/2}/(1+t)^6$ over $[0,\infty)$ by computer agrees to many decimal places. As a sanity check on size: the integrand peaks at $t=\frac{1}{3}$ with value about 0.034, and its hump has width of order 1, so a value of a few hundredths is plausible.

Common slips

- In (iv), forgetting the factor $(1+t)^{-2}$ from $\mathrm{d}x$, which gives the wrong power $p+q-2$.
- In (iii), integrating $(\sin 2\theta)^{2p-1}$ only over $[0,\frac{1}{2}\pi]$ in θ but forgetting that $\phi=2\theta$ doubles the range and halves $\mathrm{d}\theta$.
- In (v), matching p to the exponent $\frac{3}{2}$ rather than to $\frac{3}{2}+1$.

Beyond STEP

The function $\mathrm{b}(p,q)$ is Euler's beta function, usually written $B(p,q)$. It is linked to the gamma function by $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$, and the result of part (iii) is a form of Legendre's duplication formula for Γ; with $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ you can recover $3\pi/256$ in one line. The beta function also appears in probability as the normalising constant of the beta distribution. For more on such integrals, see Siklos, *Advanced Problems in Mathematics*, or Blitzstein and Hwang, *Introduction to Probability*, for the beta distribution.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.